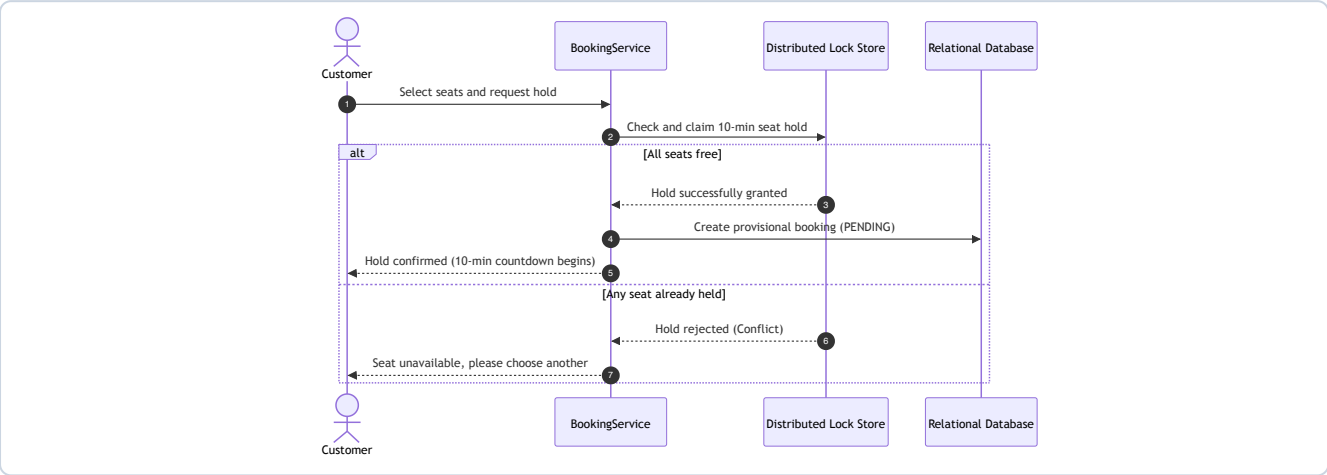


BookMyShow: Component Deep Dives

1. BookingService: Seat Locking & Idempotent Checkout

The BookingService guarantees that no two customers can reserve or purchase the same seat for any movie screening. It coordinates high-speed temporary seat reservations in a distributed lock layer, followed by transactional commitment in the database once payment succeeds.

Sequence Workflow: Distributed Seat Hold & Checkout



Step-by-Step Algorithm: Temporary Seat Hold Flow

- Step 1 (Availability Check):** The customer selects target seats on the screen layout. The system instantly verifies against the lock store that none of the requested seats are currently held or permanently booked.
- Step 2 (Atomic Hold Placement):** If all seats are unreserved, the system places a temporary 10-minute hold on the entire set of seats under the customer's active session. This operation is all-or-nothing: either all chosen seats are held together, or none are.
- Step 3 (Immediate Conflict Handling):** If even a single seat is already claimed by another user, the entire request is rejected immediately with a friendly conflict notice, allowing the customer to quickly choose alternative seats.
- Step 4 (Provisional Session Creation):** Once the temporary hold is locked, a pending booking order is recorded with a strict 10-minute expiration timestamp, and checkout proceeds to payment.
- Step 5 (Automatic Expiration):** If the customer abandons the checkout or payment is not completed before the 10-minute timer runs out, the temporary hold automatically dissolves, and the seats instantly return to the public pool without manual administrative intervention.

Idempotent Booking Protection

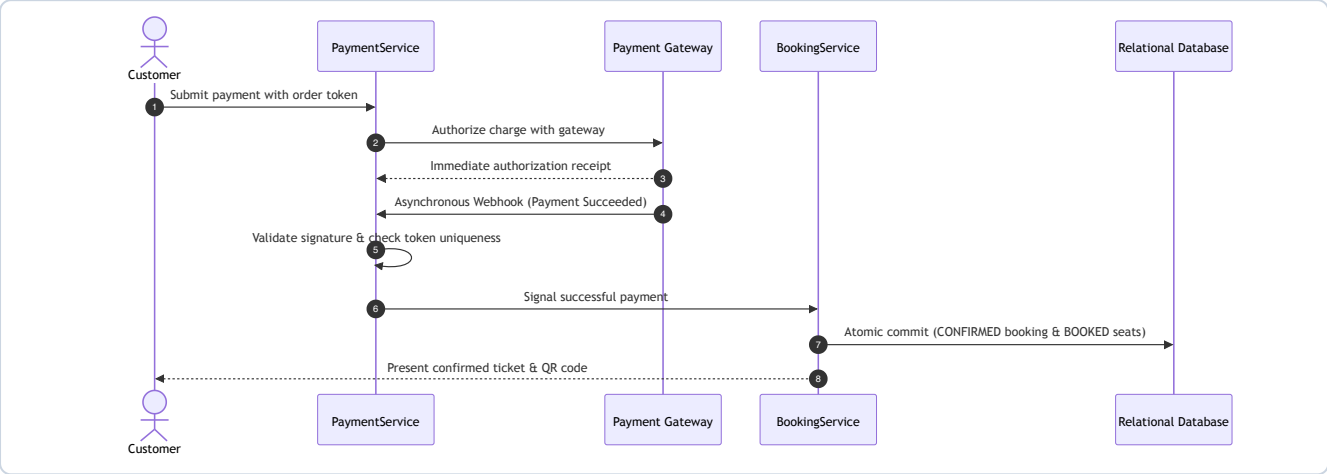
- Client-Generated Request Tokens:** Every checkout attempt includes a unique transaction token generated when the user enters the review screen. Redundant clicks on the "Pay" button re-use this token, preventing duplicate orders.
- Database Concurrency Safeguard:** The database enforces a unique composite rule on each seat for a given screening. Even under hypothetical split-second lock contention, the database hardware rejects any second write attempt outright.

BookMyShow: Component Deep Dives

2. PaymentService: 4-Stage Resilient Transaction Workflow

The PaymentService orchestrates financial exchanges with external payment processors (such as Stripe or Razorpay). It ensures zero duplicate customer charges, manages asynchronous webhook callbacks, and maintains an auditable double-entry ledger.

Sequence Workflow: Asynchronous Payment Authorization & Confirmation



Step-by-Step Algorithm: End-to-End Payment Processing Flow

- Step 1 (Checkout Initialization):** The service validates the active seat hold, calculates the final ticket amount including taxes and discounts, and registers a unique transaction token with the external payment gateway.
- Step 2 (Customer Payment Submission):** The customer provides payment credentials directly through an encrypted, compliant gateway frame. The gateway performs multi-factor customer authorization.
- Step 3 (Cryptographic Webhook Verification):** When the payment gateway completes the charge, it issues an asynchronous callback notification to the platform. The PaymentService verifies the digital signature of the callback to confirm it originates authentically from the gateway provider.
- Step 4 (Idempotent Transaction Check):** Before updating records, the service checks its payment ledger for the unique transaction token. If this token was already processed (such as from a duplicate network callback), the duplicate is safely ignored.
- Step 5 (Atomic Booking Finalization):** If the payment is confirmed, the service signals the BookingService to transition the booking status to confirmed and permanently mark the seats as booked in an atomic transaction.
- Step 6 (Failure & Refund Handling):** If the payment is declined or authorization times out, the service immediately cancels the provisional order and releases the seat hold so other customers can reserve them.

Handling External Gateway Disconnections

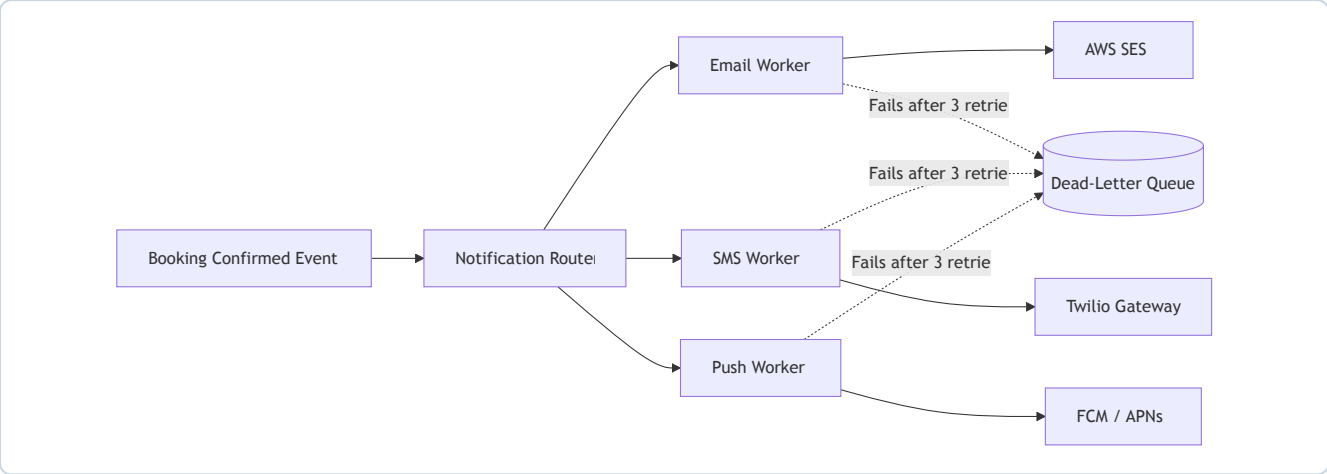
- Scheduled Reconciliation Jobs:** If an external webhook is delayed or dropped due to network issues, background reconciliation workers poll the gateway's status API every 60 seconds to resolve pending orders cleanly.
- Automatic Reversal:** If money is deducted by the bank after a customer's 10-minute reservation has expired, the reconciliation worker automatically flags the discrepancy and triggers an immediate automatic refund to the customer's account.

BookMyShow: Component Deep Dives

3. NotificationService: Multi-Channel Asynchronous Delivery

The NotificationService delivers digital tickets, booking confirmations, and showtime reminders across Email, SMS, and Mobile Push. It operates completely asynchronously via message streaming topics to prevent communication delays from slowing down customer checkouts.

Multi-Channel Dispatch & Dead-Letter Queue Architecture



Step-by-Step Algorithm: Reliable Notification Delivery Flow

- Step 1 (Event Consumption):** Worker pools continuously listen to the confirmed booking event stream. Each incoming event contains user contact details, movie metadata, and auditorium seat assignments.
- Step 2 (Deduplication Check):** The worker inspects a centralized delivery log using the booking reference. If a confirmation message has already been dispatched for this booking, the duplicate event is acknowledged and discarded.
- Step 3 (Digital Ticket & Media Generation):** The system generates a personalized scannable QR code containing encrypted ticket metadata and populates channel-specific templates (HTML email ticket, condensed SMS text, rich push notification).
- Step 4 (External Provider Dispatch):** The message payload is sent to the designated communications provider (AWS SES for email, Twilio for SMS, and FCM for push notifications).
- Step 5 (Exponential Retry on Temporary Glitches):** If a partner provider experiences a temporary outage or rate limit, the worker pauses briefly and retries up to three times with increasing delays (first after 1 second, then 2 seconds, then 4 seconds).
- Step 6 (Dead-Letter Queue Safety Routing):** If all three retry attempts fail, the message is routed into a dedicated safety holding queue (dead-letter queue) along with the error reason, allowing operations staff to inspect and re-deliver it later while normal message processing continues unaffected.

Channel Isolation & Performance Safeguards

- Independent Channel Pools:** Email, SMS, and Push workers run in completely separate pools. A temporary external delay on SMS delivery never impacts the speed of email ticket delivery or push alerts.
- Outbound Rate Quotas:** Outbound requests adhere to strict rate limiters configured to partner gateway agreements, preventing provider throttling during massive blockbuster release spikes.